

```

}))

response = service_request.execute()
entities = response['entities']
print(json.dumps(entities,indent=4))
return 0

if __name__ == '__main__':
    main()

```

Once the credentials are established, we are simply going to invoke the REST API and, specifically, the *analyzeEntities* method. The output results can be obtained from http://opensourceforu.com/article_source_code/dec16/gml.zip.

It is interesting to see that the Natural Language API has been able to correctly identify the entities (organisations and individuals) and, in some cases, also provided information on their Wikipedia entries.

Speech API

The Speech API is used to convert a RAW audio file to its equivalent text format. The API is powerful enough to work with disturbances in the audio and has multiple optimisations inbuilt to detect the text.

The code shown below is taken from the official documentation, and the parameter that we pass to it at runtime is the audio file in RAW format. The code simply reads the audio file bytes, provides information in the request about the kind of audio and then, in a synchronous fashion, gets back the results of the audio converted to text.

```

import argparse
import base64
import json
from googleapiclient import discovery
import httplib2
from oauth2client.client import GoogleCredentials
# [START authenticating]
DISCOVERY_URL = ('https://{api}.googleapis.com/$discovery/
rest?\'
                    \'version={apiVersion}\'')
def get_speech_service():
    credentials = GoogleCredentials.get_application_
default().create_scoped(
    ['https://www.googleapis.com/auth/cloud-platform'])
    http = httplib2.Http()
    credentials.authorize(http)

    return discovery.build(
        'speech', 'v1beta1', http=http,
        discoveryServiceUrl=DISCOVERY_URL)
def main(speech_file):
    with open(speech_file, 'rb') as speech:
        speech_content = base64.b64encode(speech.read())

```

```

service = get_speech_service()
service_request = service.speech().syncrecognize(
    body={
        'config': {
            'encoding': 'LINEAR16',
            'sampleRate': 16000,
            'languageCode': 'en-US',
        },
        'audio': {
            'content': speech_content.decode('UTF-8')
        }
    })
response = service_request.execute()
print(json.dumps(response))
if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument(
        'speech_file', help='Full path of audio file to be
recognized')
    args = parser.parse_args()
    main(args.speech_file)

```

The output is shown below, and is the actual audio content that was present in the file.

```

{"results": [{"alternatives": [{"confidence": 0.98267895,
"transcript": "how old is the Brooklyn Bridge"}]}]}

```

Google Machine Learning APIs provide a wide range of APIs that developers can use today to incorporate powerful machine learning algorithms in the domains of computer vision, language analysis, sentiment analysis, audio conversion and translation. The key factors that make this a compelling list of APIs is Google's track record in machine learning—the complex machine learning models that it has built over the last decade—and the ease and accuracy of the APIs. 

References

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